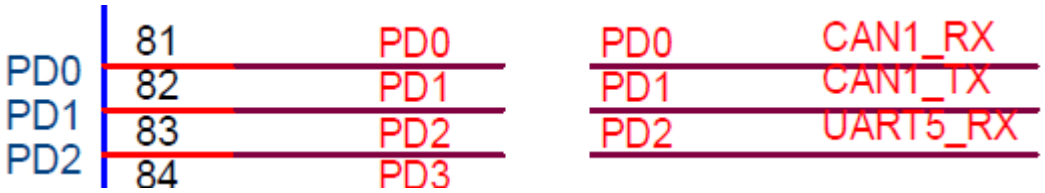
CAN(Controller Area Network) 테스트

CRZ Technology

Mango-M32F2 CAN 테스트

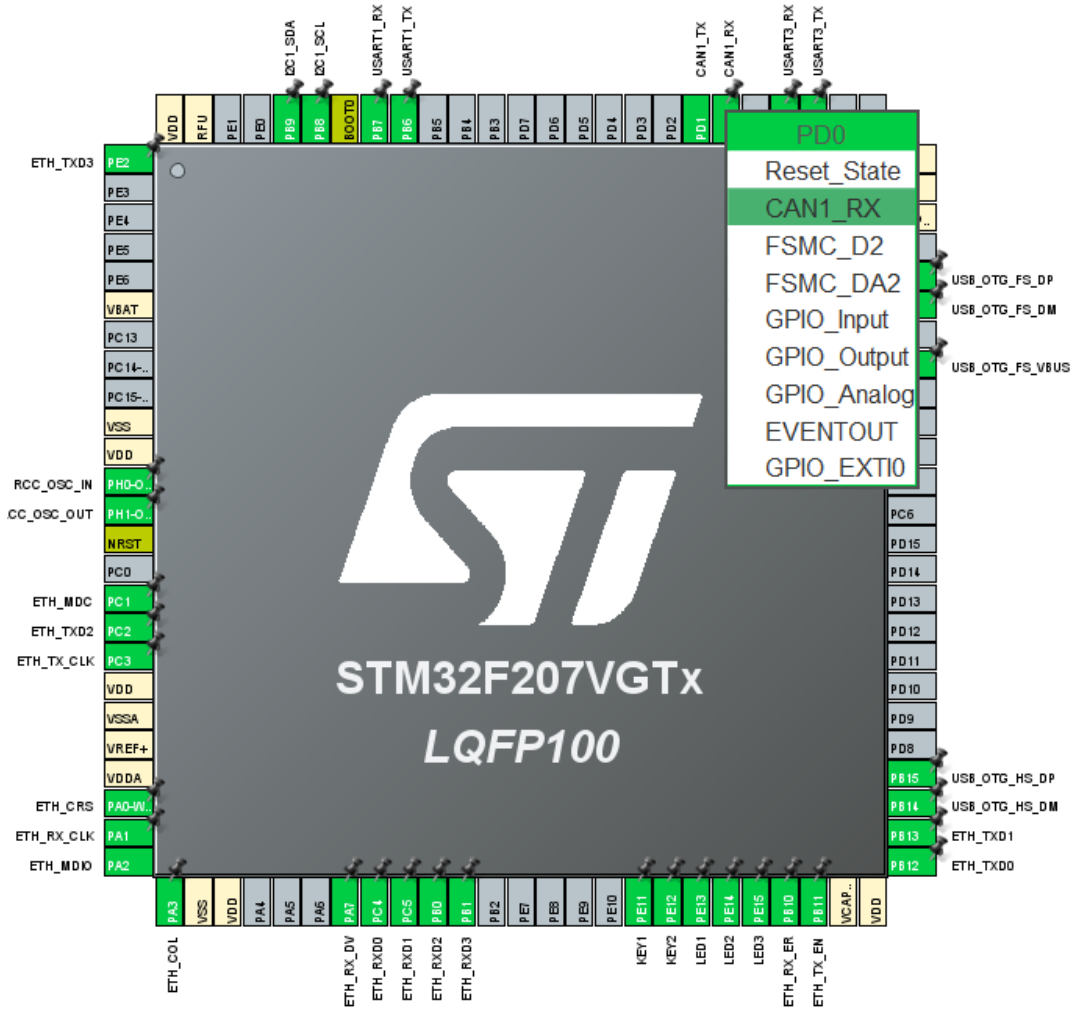
CAN pinmap

- **CAN – PD0, PD1**



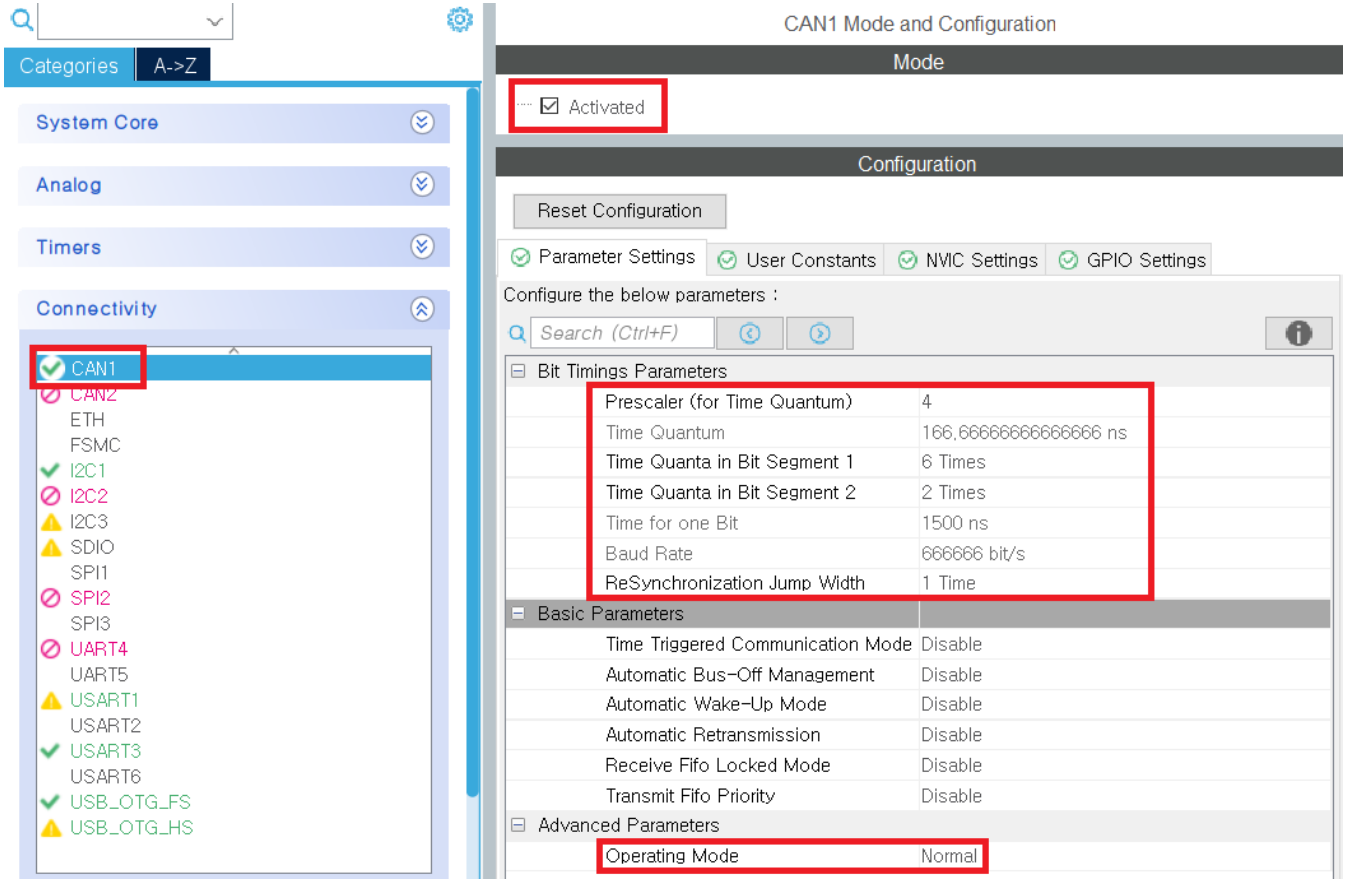
Pinout view

- **PD0-CAN1_RX, PD1-CAN1_TX 선택.**



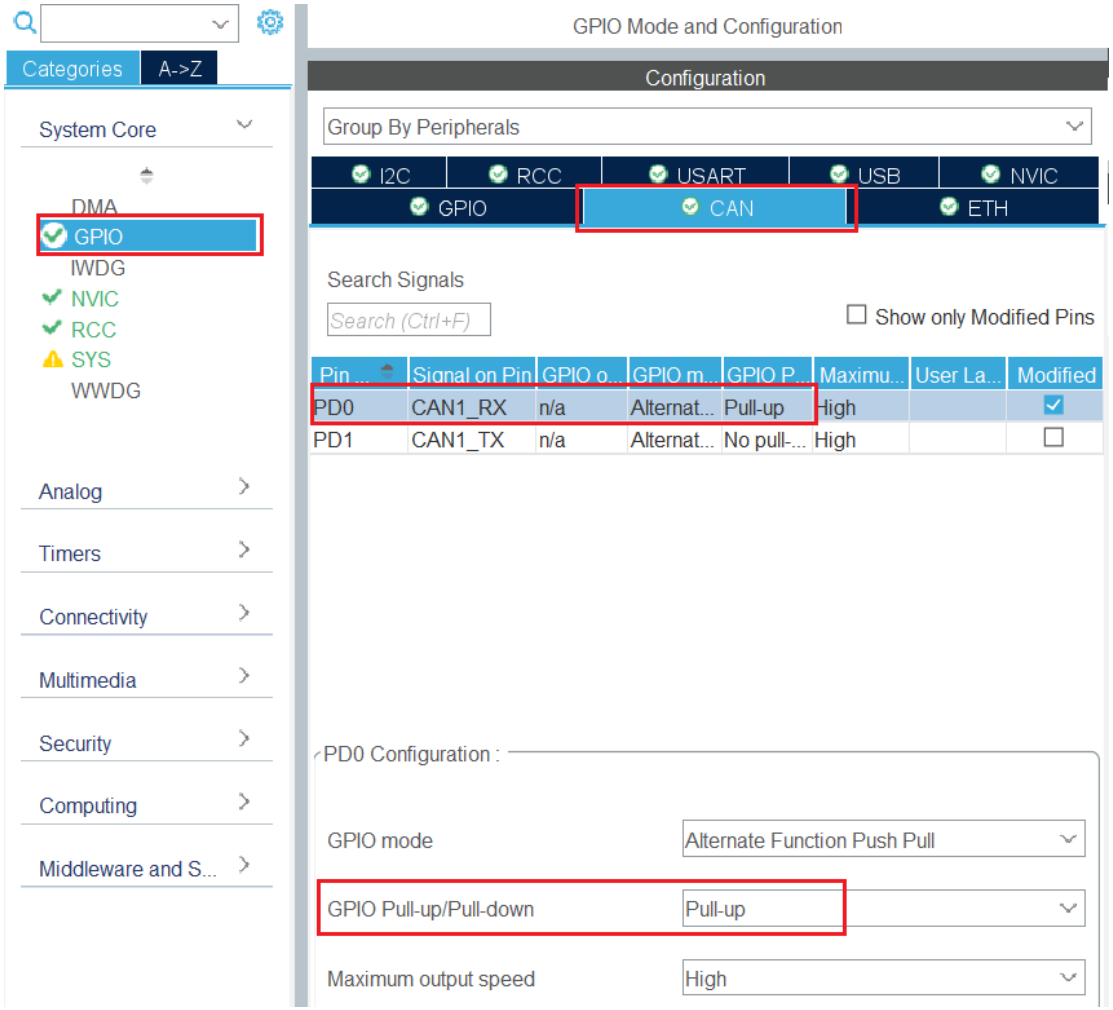
CAN 설정

- “Connectivity“ 클릭, “CAN1” 클릭.
- “Baud Rate“ - “500000”, “Operating Mode“ – “Normal Mode".



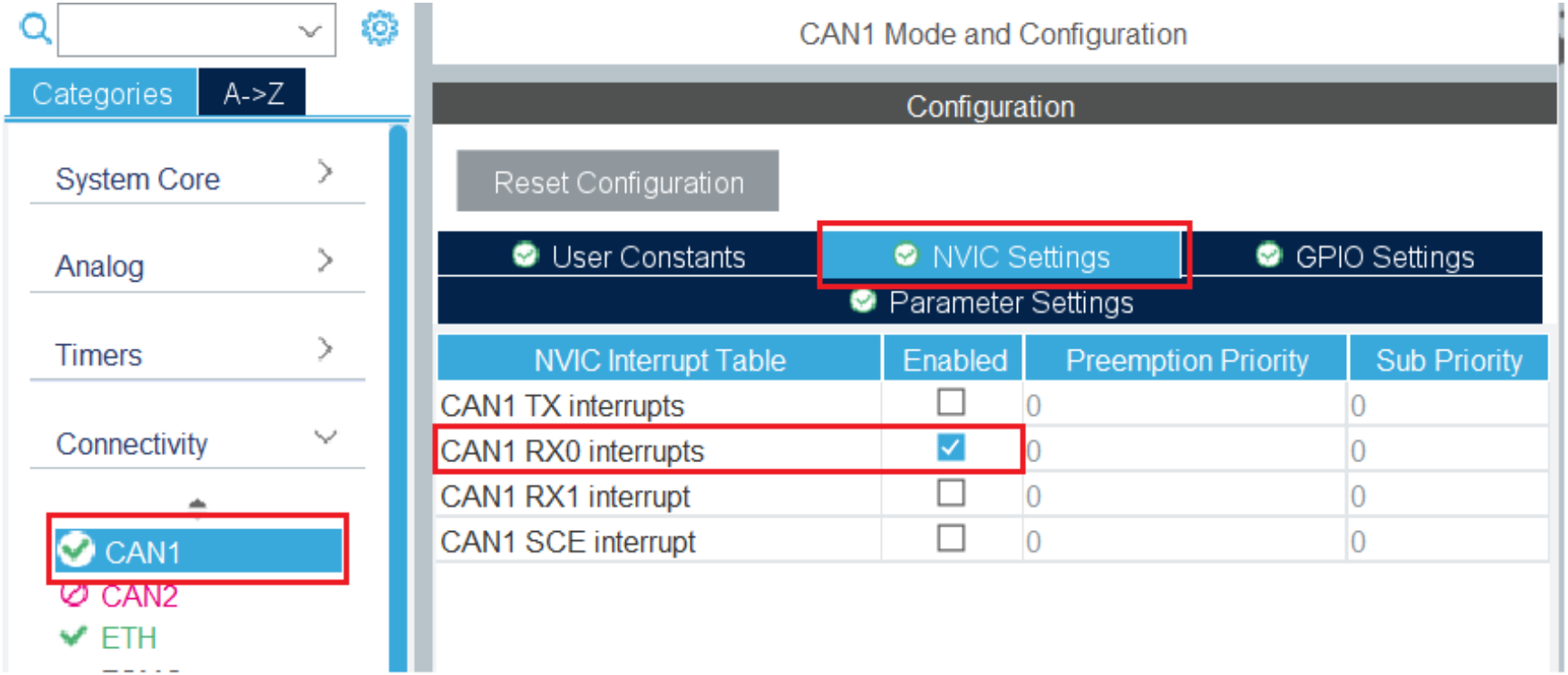
CAN 설정

- “GPIO“ 클릭, “CAN” 클릭.
- “CAN_RX“ - “Pull-up” 설정.



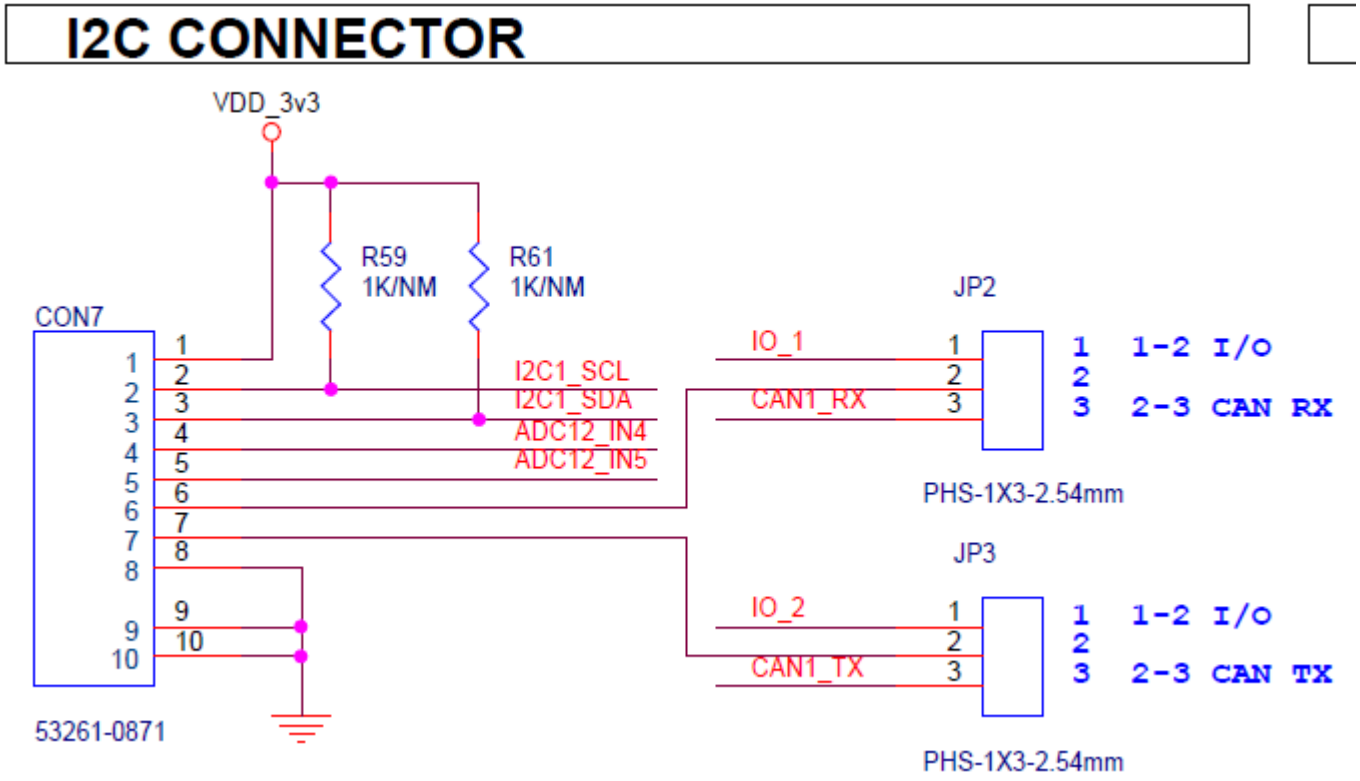
CAN 설정

- “Connectivity“ 클릭, “CAN1” 클릭.
- “CAN1 RX0 interrupt” - “Enabled”



CAN 점퍼 설정

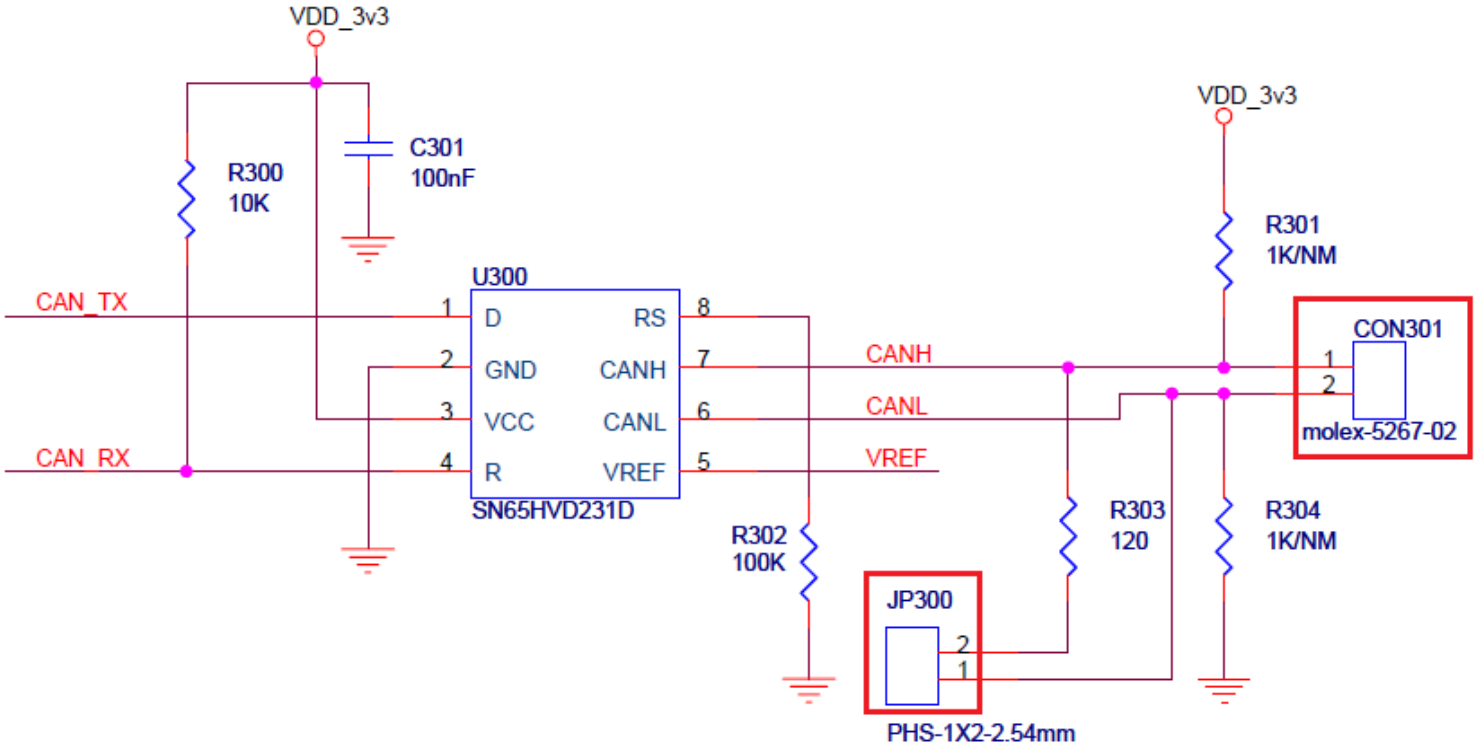
- CAN_RX – JP2 2-3 점퍼연결
- CAN_TX – JP3 2-3 점퍼연결



CX_CAN – CAN Transceiver

- CAN 동작 시 CAN Transceiver 필요.
- 종단 저항 120옴 – JP300 점퍼 연결
- 2핀 케이블을 CON301에 접속하여 2개의 CX_CAN을 서로 연결.
- 8핀 케이블을 이용, Mango-M32F2(CON7)와 CX_CAN(CON31) 연결.

SN65HVD231D



CAN 테스트

- CAN TX – KEY1 / KEY2를 누를 때 CAN 메시지 전송.
- CAN RX – CAN 메시지 수신 시 LED1 / LED2 / LED3 토글.
- TX / RX 메시지를 시리얼 터미널에 디스플레이.

CAN TX

- 송신 메시지의 ID를 0x446으로 설정

```
/* USER CODE BEGIN 2 */

HAL_CAN_Start(&hcan1);

// Activate the notification
HAL_CAN_ActivateNotification(&hcan1, CAN_IT_RX_FIFO0_MSG_PENDING);


TxHeader.DLC = 8;  // data length
TxHeader.IDE = CAN_ID_STD;
TxHeader.RTR = CAN_RTR_DATA;
TxHeader.StdId = 0x446;  // ID
/* USER CODE END 2 */
```

CAN TX

- **KEY1 / KEY2가 눌리면 CAN 메시지를 송신.**

```
void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin)
{
    if(GPIO_Pin == KEY1_Pin)
    {
        if(HAL_GPIO_ReadPin(KEY1_GPIO_Port, KEY1_Pin) == GPIO_PIN_RESET)
        {
            TxData[0] = 100;    // ms Delay
            TxData[1] = 40;     // loop rep
            TxData[2] = 0xaa;
            TxData[3] = 0xaa;
            TxData[4] = 0xaa;
            TxData[5] = 0xaa;
            TxData[6] = 0xaa;
            TxData[7] = 0xaa;

            HAL_CAN_AddTxMessage(&hcan1, &TxHeader, TxData, &TxMailbox);
            printf("TX:%02x %02x %02x %02x %02x %02x %02x %02x\r\n", TxData[0],
        }
    }
```

CAN RX

- 수신 메시지 ID를 0x446으로 필터 설정.

```
/* USER CODE BEGIN CAN1_Init 2 */
CAN_FilterTypeDef canfilterconfig;

canfilterconfig.FilterActivation = CAN_FILTER_ENABLE;
canfilterconfig.FilterBank = 18;  // which filter bank to use from the assigned ones
canfilterconfig.FilterFIFOAssignment = CAN_FILTER_FIFO0;
canfilterconfig.FilterIdHigh = 0x446<<5;
canfilterconfig.FilterIdLow = 0;
canfilterconfig.FilterMaskIdHigh = 0x446<<5;
canfilterconfig.FilterMaskIdLow = 0x0000;
canfilterconfig.FilterMode = CAN_FILTERMODE_IDMASK;
canfilterconfig.FilterScale = CAN_FILTERSCALE_32BIT;
canfilterconfig.SlaveStartFilterBank = 20;  // how many filters to assign to the CAN1 (master can)

HAL_CAN_ConfigFilter(&hcan1, &canfilterconfig);
/* USER CODE END CAN1_Init 2 */
```

CAN RX

- **RX callback을 활성화 하고 callback 함수에서 메시지를 수신.**

```
// Activate the notification
HAL_CAN_ActivateNotification(&hcan1, CAN_IT_RX_FIFO0_MSG_PENDING);
```

```
void HAL_CAN_RxFifo0MsgPendingCallback(CAN_HandleTypeDef *hcan)
{
    HAL_CAN_GetRxMessage(hcan, CAN_RX_FIFO0, &RxHeader, RxData);
    if (RxHeader.DLC == 8)
    {
        datacheck = 1;
    }
}
```

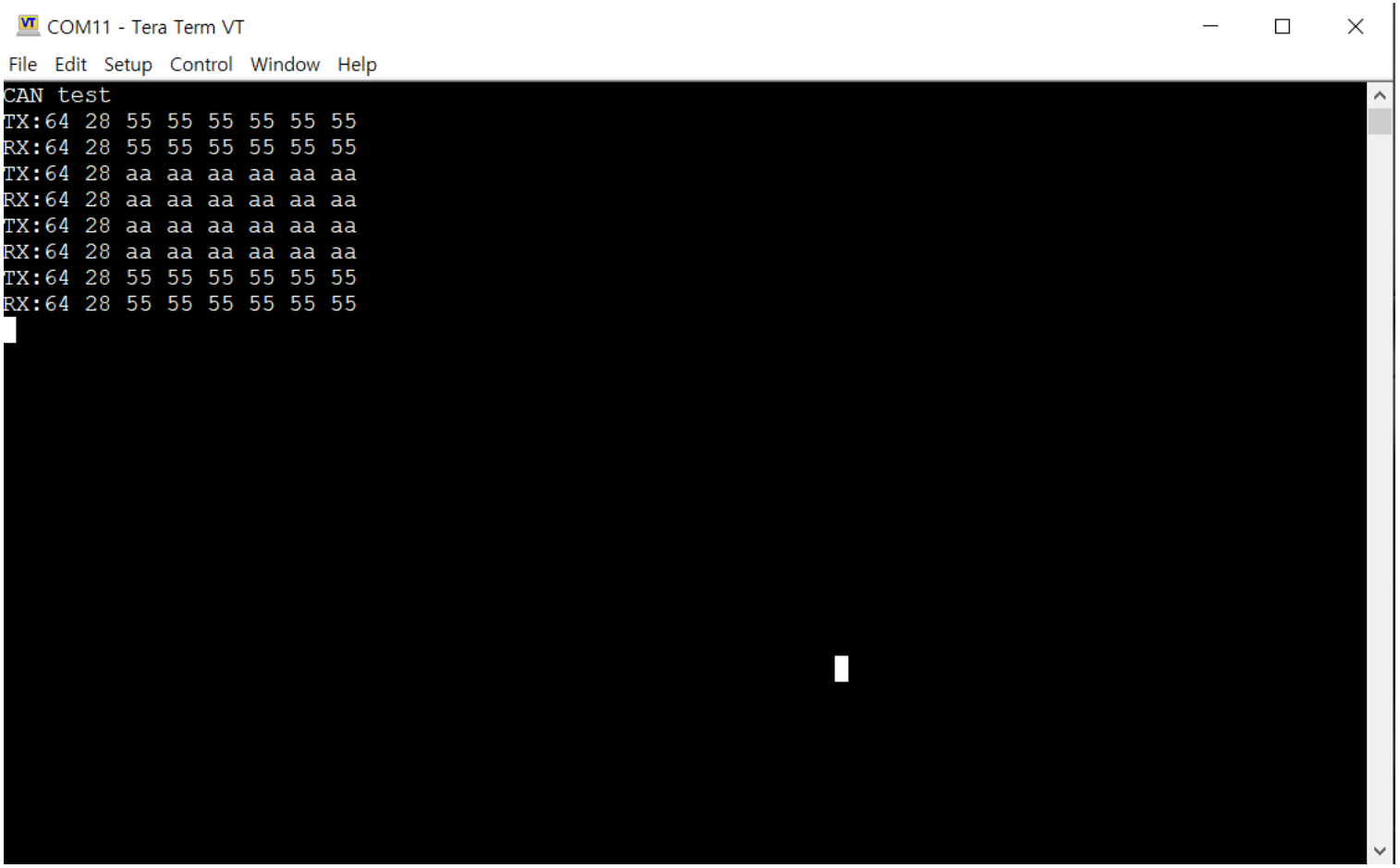
CAN RX

- **Main loop에서 수신 메시지 프린트.**

```
/* Infinite loop */
/* USER CODE BEGIN WHILE */
while (1)
{
  /* USER CODE END WHILE */

  /* USER CODE BEGIN 3 */
    if (datacheck)
    {
        printf("RX:%02x %02x %02x %02x %02x %02x %02x %02x\r\n", RxData[0],

          // blink the LED
        for (int i=0; i<RxData[1]; i++)
        {
            HAL_GPIO_TogglePin(GPIOE, LED1_Pin | LED2_Pin | LED3_Pin);
            HAL_Delay(RxData[0]);
        }

        datacheck = 0;
    }
}
/* USER CODE END 3 */
```

Test 결과 확인

- **CAN 송신 / 수신 메시지 프린트 확인.**



Thank You